Toward a Simulated Attack Dataset for Private 5G Network Security Using Testbeds*

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Abstract

For Private 5G (P5G) security, AI-based intrusion detection requires network datasets reflecting P5G network characteristics. To address this need, we design attack scenarios targeting four services on a dedicated P5G testbed and collect a network dataset for AI-based intrusion detection in P5G environments.

* Proceedings of The 2026 IFIP WG 8.4 International Symposium on E-Business Information Systems Evolution (EBISION 2026), Article No. P-39, July 8-10, 2026, Seoul, South Korea.
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1 Attack Scenarios

To collect realistic attack traffic, we identified four target services commonly found in P5G enterprise deployments and conducted penetration tests on each service within the P5G testbed. The attack scenarios and tools used for each service are summarized in Table 1.

Table 1: Attack scenarios per service

Target Service	Attack Type	Severity	Primary Tools
IoT (MQTT)	Information Gathering	High	Password Cracker, Packet Crafting Tool,
	SYN & ACK Flooding, Brute Force	Critical	
Web (Nginx)	Reconnaissance & Vulnerability Scanning, Path Traversal	High	Web Vulnerability Scanner, DoS Tool, Exploitation Framework
	Slow HTTP DoS, Exploit Attempt	Critical	
Media Streaming	Service Identification	High	DoS Tool, Proxy-Based Request Tool
	Slow HTTP DoS, Connection Resource Exhaustion	Critical	
DB (MariaDB)	Service Identification, Proxy-based Access Bypass	High	Password Brute-Forcer, Proxy Chaining Tool, SQL Injection Tool, Exploitation Framework
	Brute Force, SQL Injection, Authentication Bypass	Critical	

Among the tested services, Denial of Service (DoS) availability attacks had the largest service-level impact under the tested conditions. The media streaming service showed severe response degradation under slow-connection flooding, and the IoT broker was similarly sensitive to TCP SYN and MQTT CONNECT flooding. The database server was protected by network-level access controls that blocked external connections to the backend, so brute-force, SQL injection, and authentication-bypass traffic was collected at the network boundary rather than at the server. No code execution or privilege escalation was observed.

2 P5G Testbed and Data Collection

Although attacking an operational P5G network would yield the most realistic data [1], [2], it is not practical due to service disruption risks and exposure of subscriber data. Hence, as shown in Figure 1, we built a dedicated P5G testbed that integrates a P5G segment with a legacy enterprise network.

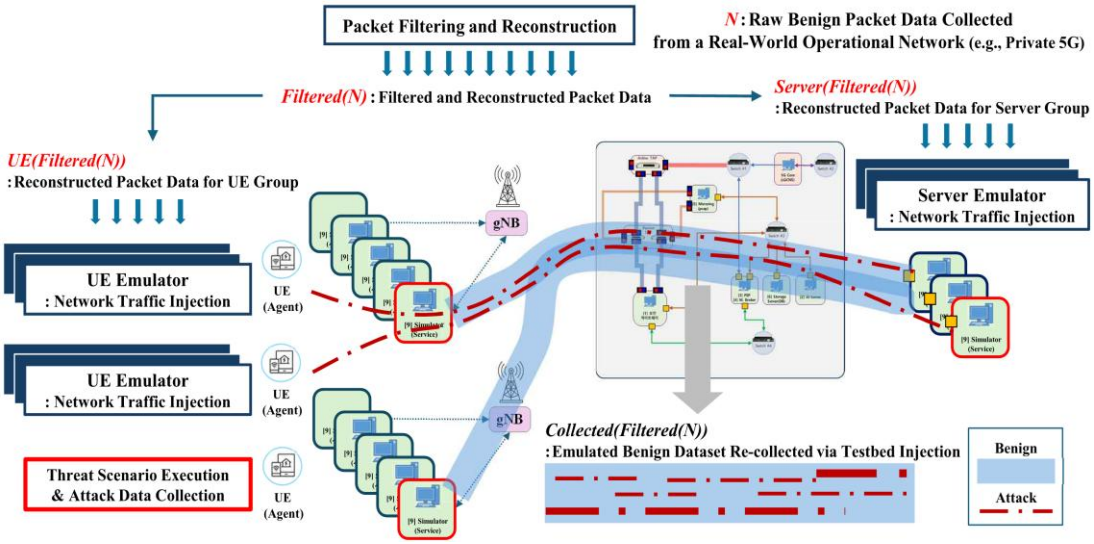


Figure 1: Test Network for the Integrated Environment of Private 5G

The testbed comprises a P5G core (UPF, AMF, SMF, etc.), multiple next generation Node Bs (gNBs) and User Equipment (UEs), and a legacy enterprise network for the target services. Each component was configured to replicate the settings of an operational P5G deployment. The P5G core and the radio configuration follow an operational P5G setting, while the enterprise services and attack execution were reproduced through emulator replay rather than live attacker UEs.

To address the limitations of widely used datasets such as NSL-KDD [3] and CICIDS2017 [4] in representing real-world traffic, we collect traffic data in the testbed of P5G. For this, we replay the captured attack traffic through a UE emulator and a server emulator, allowing flexible variation of transmission schedules, packet rates, attack intensity, and the number of concurrently active UEs. The replayed attack traffic is then mixed with benign traffic generated by other UEs so that each capture contains both benign and attack traffic under the same radio and core conditions. During collection, attack and benign traffic were generated from separate UEs toward different destination hosts, and each attack tool was run in a separate, logged time window. Each flow is labeled by matching its 5-tuple and timestamp against the logged attack metadata. The collected dataset serves as a foundation for evaluating P5G intrusion detection models. As a result, we collected approximately 62GB of network traffic over several days, comprising 91% benign traffic and 9% attack traffic.

In future work, we plan to extend coverage to 5G-specific attacks at the signaling (NAS, PFCP) and user-plane (GTP-U) layers, and to use the dataset both to validate AI-based intrusion detection models [5] and as ground-truth data for generative AI-based traffic synthesis [6]. Finally, our study will help to enhance the capabilities for P5G-based E-Business security.

3 Acknowledgments

This work was supported by the Institute of Information & Communications Technology Planning & Evaluation (IITP) grant funded by the Korea government (MSIT) (RS-2024-00397469, Development of private 5G Security Technology for Integrated Private 5G and Enterprise Network Security)

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